

Applied Linear Algebra

NumPy and PyTorch

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Try the following in a Python shell

```
1      import numpy as np
2
3      a = np.array([1, 2, 3])
4
5      assert(a.dtype == np.int64)
6
7      # what is the shape of 'a'?
8      assert(a.shape == ?)
9
```

Array Shape

NumPy

```
1  import numpy as np
2
3  a = np.array([1, 2, 3])
4  assert(a.shape == (3,))
5
6  assert(a.shape != (3)) # why?
7  assert(a.shape != 3) # why so?
8
9  ## what's the type of a.shape?
10 assert(isinstance(a.shape, ?))
```

Array of Zeroes

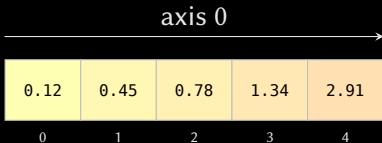
NumPy

```
1  import numpy as np
2
3  a = np.zeros(4)
4  assert a == [0, 0, 0, 0] # what's up with is?
5
6  # 'array_equal'
7  assert np.array_equal(a, [0, 0, 0, 0])
8
9  # explain this!
10 assert (a == np.array([0, 0, 0, 0])).all()
11
12 # what's the type of a.shape?
13 assert(isinstance(a.shape, ?))
```

1-Dimensional ndarray

NumPy

```
1 import numpy as np
2
3 a = np.array([0.12, 0.45, 0.78, 1.34, 2.91])
4 assert(a.shape == (5,))
5 assert(a.ndim == 1)
6 assert(a.size == 5)
```



2-Dimensional ndarray

NumPy

```
1  # A 3x5 2D ndarray/tensor
2  a = np.array([
3      [0.12, 0.45, 0.78, 1.34, 2.91],
4      [0.23, 0.56, 0.89, 1.45, 3.02],
5      [0.34, 0.67, 0.90, 1.56, 3.13],
6  ])
7  assert(a.shape == (3, 5))
8  assert(a.ndim == 2)
```

The diagram illustrates a 2D array with 3 rows and 5 columns. The vertical axis is labeled 'axis 0' with a downward arrow, and the horizontal axis is labeled 'axis 1' with a rightward arrow. The columns are indexed 0 to 4, and the rows are indexed 0 to 2. The values are displayed in a grid where the first column is light red, the second is light blue, and the remaining three are light green.

		axis 1 →				
		0	1	2	3	4
axis 0 ↓	0	0.12	0.45	0.78	1.34	2.91
	1	0.23	0.56	0.89	1.45	3.02
	2	0.34	0.67	0.90	1.56	3.13

3-Dimensional ndarray

NumPy

```
1 import numpy as np
2
3 # A 3x3x4 3D tensor: shape (planes, rows, cols)
4 a = np.random.rand(3, 3, 4)
5 assert(a.shape == (3, 3, 4))
6 assert(a.ndim == 3)
```

